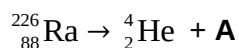


2015 H2 Prelim Paper 1 Suggested Solutions

1	Consider the following half-equations. $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$ $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$ Which of the following reflects correctly the volume of $0.100 \text{ mol dm}^{-3}$ of $\text{K}_2\text{Cr}_2\text{O}_7$ and 0.15 mol dm^{-3} of FeI_2 required for complete reaction?		
		Volume of $\text{K}_2\text{Cr}_2\text{O}_7$	Volume of FeI_2
A	12.50 cm^3	25 cm^3	
B	12.50 cm^3	50 cm^3	
C	18.75 cm^3	25 cm^3	
D	31.25 cm^3	50 cm^3	
Answer: C $\text{FeI}_2 \rightarrow \text{Fe}^{2+} + 2\text{I}^-$ Amt of $\text{FeI}_2 = (25/1000) \times 0.15 = 0.00375 \text{ mol}$ 7.5×10^{-3} Amt of $\text{Fe}^{2+} = 0.00375 \text{ mol}$ Amt of $\text{I}^- = 0.0075 \text{ mol}$ $6\text{Fe}^{2+} \equiv \text{Cr}_2\text{O}_7^{2-}$ $6\text{I}^- \equiv \text{Cr}_2\text{O}_7^{2-}$ Amt of $\text{Cr}_2\text{O}_7^{2-}$ reacted with $\text{Fe}^{2+} = 0.000625 \text{ mol}$ 1.25×10^{-3} Amt of $\text{Cr}_2\text{O}_7^{2-}$ reacted with $\text{I}^- = 0.00125 \text{ mol}$ 3.75×10^{-3} Total amt of $\text{Cr}_2\text{O}_7^{2-} = 0.001875 \text{ mol}$ Volume of $\text{Cr}_2\text{O}_7^{2-}$ required = $0.001875 / 0.1 = 0.01875 \text{ dm}^3 = 18.75 \text{ cm}^3$			

- 2** Long term exposure to radioactive isotope $^{226}_{88}\text{Ra}$ increases the risk of developing diseases such as bone cancer and leukaemia.

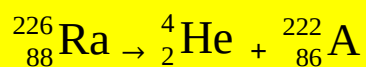
$^{226}_{88}\text{Ra}$ decays to give an element **A** and emits a high energy α -particle (which is a helium nucleus, ^4_2He). No other particle is produced. α -Particles cause irreparable damage to the tissues of internal organs.



Which row in the table correctly describes the nuclear make-up of $^{226}_{88}\text{Ra}$ and element **A**?

	$^{226}_{88}\text{Ra}$	A	
	number of neutrons	number of protons	number of neutrons
A	138	84	136
B	138	86	136
C	226	84	138
D	226	86	138

Answer: B



- 3** Beams of charged particles are deflected by an electric field. In an experiment, protons are deflected through an angle of $+15^\circ$.

A beam of particles, **F**, are separately passed through the same electric field and was deflected by an angle of $+5^\circ$. Each **F** particle has 12 times the mass of a proton.

What is the overall charge on a particle of **F**?

A	1+
B	2+
C	3+
D	4+

Answer: D

e/m of proton: $1/1 = 1$ angle: 15°

to have angle of 5° , e/m of **F** is $1/3$. Since mass of **F** is 12, charge is 4+

4	Which pair of compounds below satisfies the following conditions? (i) The first compound has a larger bond angle than the second compound. (ii) The first compound is more polar than the second compound.			
		First compound	Second compound	
	A	ClO_2	HCN	
	B	NF_3	SeF_6	
	C	IF_3	PH_3	
	D	BeCl_2	N_2H_4	
Answer: B				
A		ClO_2 (105° , polar)	HCN (180° , polar)	
B		NF_3 (107°, polar)	SeF_6 (90°, non-polar)	
C		IF_3 (90° , polar)	PH_3 (107° , polar)	
D		BeCl_2 (180° , non-polar)	N_2H_4 (107° , polar)	

5	The theoretical magnitude of lattice energy can be estimated from the Born-Landé equation, the relationship of which can be simplified as shown. $ \text{lattice energy} \propto \left \frac{q_+ \times q_-}{r_+ + r_-} \right $ The consequence of the Born-Landé equation is that aluminium oxide has a higher theoretical magnitude of lattice energy as compared to magnesium oxide. However, the melting point of aluminium oxide is 2070 °C as compared to 2850 °C for magnesium oxide. Which of the following could not be a possible reason for this anomaly?		
	A	Aluminium oxide has a different lattice structure (i.e. different pattern of ions arrangement) as compared to magnesium oxide.	
	B	Aluminium oxide has a more significant covalent character as compared to magnesium oxide.	
	C	Aluminium ion has a much higher charge density as compared to magnesium ion.	
	D	Aluminium has a stronger metallic bond as compared to magnesium.	
Answer: D			
(A) Different mole ratio of cations to anions, as well as different relative radii of cations to anions will result in different lattice structure for different ionic compounds. This will result in different magnitude of lattice energy.			
(B) Al^{3+} has very high charge density which will polarise O^{2-} ion to a larger extent than in MgO . This increases covalent character of Al_2O_3 , which decreases its ionic bond strength, its magnitude of lattice energy and its melting point.			
(C) charge density $\propto \frac{q_+}{r_+}$ Mg^{2+} and Al^{3+} has similar ionic radius, but Al^{3+} has higher ionic charge. Hence, Al^{3+} has higher charge density which polarises O^{2-} ion more and decreases magnitude of lattice energy.			
(D) Aluminium metal has stronger metallic bond than magnesium metal. However, this does not affect the properties of Al_2O_3 and MgO which are ionic compound.			

6	Which of the following statements best explains why iodine is a solid at room temperature but water is a liquid?	
	A	Iodine has a much larger molar mass than water.
	B	Iodine has strong and extensive covalent bonds between the atoms in a giant 3D structure.
	C	The van der Waals' forces of attraction between iodine molecules are stronger than the intermolecular hydrogen bonding of water.
	D	Water has intermolecular hydrogen bonding which are stronger than the van der Waals' forces of attraction between iodine molecules.

Answer: **C**

This implies that **water has a lower melting point than iodine**.

Option A is true BUT does not explain why iodine has a higher melting point.

Option B is false because **iodine has a simple molecular structure**, NOT giant molecular.

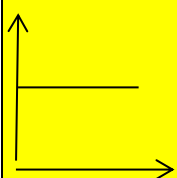
Option C is true. Melting involves overcoming the **intermolecular forces of attraction**.

Option D is false because this contradicts with the observation that water has a lower melting point.

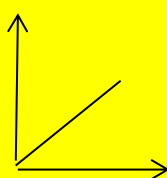
There are van der Waals' forces of attractions between iodine molecules and hydrogen bonding exists between water molecules. **While we normally describe van der Waals' forces as weak, but because of the very much larger size of electron cloud in this case, the van der Waals' forces are actually stronger than the hydrogen bonding in water.**

7	For a given mass of an ideal gas, which graph displays a different shape from the rest?			
	A	PV against $\frac{1}{V}$ (at constant T)	C	$\frac{\text{density}}{\text{pressure}}$ against P (at constant T)
	B	$\frac{V}{T}$ against T (at constant P)	D	$\frac{1}{\text{density}}$ against T (at constant P)

Answer: **D**



graphs: A, B & C



graph: D

MCQ technique: If one side consists of two variable, it is usually a horizontal line. (DO Not use this in paper 2 and 3!!)

8	For the reaction $2\text{B}(\text{aq}) + \text{C}(\text{aq}) \longrightarrow \text{D}(\text{aq})$ at room temperature, the rate equation is $\text{Rate} = k[\text{H}^+][\text{B}]^2$ Which statement is not correct?
A	H^+ is a catalyst in the reaction.
B	When the concentration of C is halved at 15°C , the rate decreases.
C	When the concentrations of H^+ and B are doubled, the rate increases six times.
D	The slow step of the reaction could be $2\text{B}(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{E}(\text{aq})$, where E is an intermediate.
Answer: C When $[\text{H}^+]$ is doubled, rate increases by 2 times. When $[\text{B}]$ is doubled, rate increases by 4 times. $\Rightarrow$ Overall rate increases by $2 \times 4 = 8$ times.	

9	The pH of water varies with temperature according to the table below. <table border="1"> <thead> <tr> <th>Temperature / $^\circ\text{C}$</th><th>pH</th></tr> </thead> <tbody> <tr> <td>25</td><td>7.0</td></tr> <tr> <td>62</td><td>6.5</td></tr> </tbody> </table> Which of the following statements is true?	Temperature / $^\circ\text{C}$	pH	25	7.0	62	6.5
Temperature / $^\circ\text{C}$	pH						
25	7.0						
62	6.5						
A	The K_a is higher than K_b at 62°C .						
B	Water becomes more acidic as temperature increases.						
C	The ionic dissociation of water is an exothermic process.						
D	The ionic product of water, K_w , increases at higher temperature.						
Answer: D A is not true. Since $K_w = K_a \times K_b$, They have the same value. $K_w = [\text{H}^+][\text{OH}^-]$, hence $[\text{H}^+] = [\text{OH}^-] \rightarrow$ B not true. C is not true. As temperature increases, pH is lowered, indicating more $[\text{H}^+]$. Eqm position has shifted to the right favouring the endothermic reaction. D is true. At 62°C, $K_w = [\text{H}^+][\text{OH}^-] = [\text{H}^+]^2 = (10^{-6.5})^2 = 10^{-13} \text{ mol}^2\text{dm}^{-6}$.							

10	The use of the <i>Data Booklet</i> is required for this question. A voltaic cell is made up of Mg^{2+}/Mg half-cell and $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cell. Which one of the following statements is correct?
	A The Mg^{2+}/Mg half-cell is the positive electrode.
	B Increasing the temperature has no effect on the e.m.f. of the cell.
	C Addition of water to the $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cell decreases the e.m.f. of the cell.
	D Addition of aqueous sodium hydroxide to the Mg^{2+}/Mg half-cell decreases the e.m.f. of the cell.

Answer: B

$\text{Mg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mg}$ $E^\ominus = -2.38\text{V}$ ($E^\ominus$ less positive, [O] occurs at this half-cell) $\rightarrow$ anode (negative terminal)

$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$ $E^\ominus = +0.77\text{V}$ ($E^\ominus$ more positive, [R] occurs at this half-cell) $\rightarrow$ cathode (positive terminal)

A is not true.

B is true. Temperature has no effect on both half-cell reactions.

C is not true. Addition of water dilutes the Fe ions equally, eqm position does not shift.

D is not true. As $\text{Mg}(\text{OH})_2$ ppt is formed, eqm position of $\text{Mg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mg}$ shifts to the left to favour oxidation. E of Mg^{2+}/Mg becomes more negative, thus e.m.f. decreases.

11	20 cm ³ of 0.2 moldm ⁻³ of H ₂ SO ₄ was mixed with 20 cm ³ of 0.2 moldm ⁻³ of NaOH. The reaction mixture increased by x °C. Which of the following would give a similar temperature rise?
A	10 cm ³ of 0.4 moldm ⁻³ of H ₂ SO ₄ was mixed with 20 cm ³ of 0.1 moldm ⁻³ of NaOH.
B	20 cm ³ of 0.2 moldm ⁻³ of H ₂ SO ₄ was mixed with 20 cm ³ of 0.4 moldm ⁻³ of NaOH.
C	10 cm ³ of 0.1 moldm ⁻³ of H ₂ SO ₄ was mixed with 10 cm ³ of 0.4 moldm ⁻³ of NaOH.
D	20 cm ³ of 0.1 moldm ⁻³ of H ₂ SO ₄ was mixed with 20 cm ³ of 0.1 moldm ⁻³ of NaOH.

Answer: C

$$n_{H_2O} = 0.004$$

$$c_{H_2O} = \underline{\underline{0.100 \text{ moldm}^{-3}}}$$

$$\text{A: } c_{H_2O} = 0.0667 \text{ moldm}^{-3}$$

$$\text{B: } c_{H_2O} = 0.400 \text{ moldm}^{-3}$$

$$\text{C: } c_{H_2O} = \underline{\underline{0.100 \text{ moldm}^{-3}}}$$

$$\text{D: } c_{H_2O} = 0.010 \text{ moldm}^{-3}$$

OR

	Limiting Reagent	n_{H_2O}	$V_{\text{total}} / \text{cm}^3$	$\frac{n_{H_2O}}{V_{\text{total}}} / 10^{-4}$
Qn	NaOH	0.004	40	1
A	NaOH	0.002	30	0.667
B	Both	0.008	40	2
C	H ₂ SO ₄	0.002	20	1
D	NaOH	0.002	40	0.5

12	Dinitrogen oxide burns in ethyne to produce water vapour, carbon dioxide and nitrogen gas as the only products. $5\text{N}_2\text{O}(\text{g}) + \text{C}_2\text{H}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g}) + 2\text{CO}_2(\text{g}) + 5\text{N}_2(\text{g})$ Given that the N=N and N=O bond energies in dinitrogen oxide is $+418 \text{ kJ mol}^{-1}$ and $+686 \text{ kJ mol}^{-1}$ respectively, what is the enthalpy change of the reaction?		
	<table border="1"> <tr> <td data-bbox="260 477 300 544">A</td><td data-bbox="308 477 1426 544">$+1670 \text{ kJ mol}^{-1}$</td></tr> </table>	A	$+1670 \text{ kJ mol}^{-1}$
A	$+1670 \text{ kJ mol}^{-1}$		
	<table border="1"> <tr> <td data-bbox="260 544 300 611">B</td><td data-bbox="308 544 1426 611">$+1710 \text{ kJ mol}^{-1}$</td></tr> </table>	B	$+1710 \text{ kJ mol}^{-1}$
B	$+1710 \text{ kJ mol}^{-1}$		
	<table border="1"> <tr> <td data-bbox="260 611 300 678">C</td><td data-bbox="308 611 1426 678">$-1670 \text{ kJ mol}^{-1}$</td></tr> </table>	C	$-1670 \text{ kJ mol}^{-1}$
C	$-1670 \text{ kJ mol}^{-1}$		
	<table border="1"> <tr> <td data-bbox="260 678 300 745">D</td><td data-bbox="308 678 1426 745">$-1710 \text{ kJ mol}^{-1}$</td></tr> </table>	D	$-1710 \text{ kJ mol}^{-1}$
D	$-1710 \text{ kJ mol}^{-1}$		
Answer: C N_2O structure: $\text{N}=\text{N}=\text{O}$ $\Delta H = 5 \times 418 + 5 \times 686 + 840 + 2 \times 410 - (2 \times 460 + 4 \times 740 + 5 \times 994)$ $= -1670 \text{ kJ mol}^{-1}$			

13	Two elements G and H have the following properties.		
	G and H form ionic compounds with the formula of CaG and CaH_2 respectively. Element G forms GH_6 molecule while H does not form HG_6 .		
	Which pair of electronic configuration of G and H is correct?		
	(G and H do not represent elements in the periodic table)		
		G	H
	A $[\text{He}] 2s^2 2p^4$	$[\text{He}] 2s^2 2p^2$	
	B $[\text{He}] 2s^2 2p^5$	$[\text{Ne}] 3s^2 3p^2$	
	C $[\text{Ne}] 3s^2 3p^5$	$[\text{He}] 2s^2 2p^5$	
	D $[\text{Ne}] 3s^2 3p^4$	$[\text{He}] 2s^2 2p^5$	
Answer: D			
G is sulfur from Gp VI and H is fluorine from Gp VII to form compounds of CaS and CaF_2 respectively.			
G can expand octet while H cannot $\rightarrow$ Period 3 and below.			
Hence G is $[\text{Ne}] 3s^2 3p^4$ (Period 3, Gp VI) and H is $[\text{He}] 2s^2 2p^5$ (Period 2, G VII)			

14	Which of the following statement is incorrect?
A	Metal ions in aqueous solution behave as Lewis acids.
B	Beryllium ion undergoes hydration and partial hydrolysis in water.
C	As the size of the cation decreases, the magnitude of enthalpy change of hydration increases.
D	The equilibrium constant for the hydrolysis of a hydrated cation is analogous to the K_a for the ionisation of a weak monoprotic acid.

Answer **B**

A: Metal ions accept electron pairs from water, acting as Lewis acids.

B: Beryllium is analogous to aluminium and undergoes complete hydrolysis. Option B is incorrect.

C: $|\Delta H_{\text{hyd}}| \propto \left| \frac{q_+}{r_+} \right|$, as $r_+ \downarrow$, $|\Delta H_{\text{hyd}}| \uparrow$

D:

Hydrolysis of cations : $[M(H_2O)_n]^{x+} \rightleftharpoons [M(H_2O)_{n-1}(OH)]^{(x-1)+} + H^+$

$$K_{\text{eqb}} = \frac{[H^+][M(H_2O)_{n-1}(OH)]^{+(x-1)}}{[M(H_2O)_n]^{+x}}$$

Ionisation of weak acid : $HA \rightleftharpoons H^+ + A^-$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

Note the similarity of the equation for the hydrolysis of a hydrated cation with the equation for the ionization of a weak acid in aqueous solution.

15	J, K and L are 3 consecutive elements in period 3 of the periodic table. The following information are given on the elements: • J forms a cation. • K has a higher melting point than J. • L forms an anion. Which of the following statement accounts for the above information?
A	Electrical conductivity increases from J to L .
B	The cationic radius of J is smaller than that of the cationic radius of K .
C	The number of electrons in the cation of J is more than the cation of K .
D	Effervescence is seen when a magnesium strip is dipped into the aqueous solution containing the chloride of element L .

Answer: **D**

since J, K and L are consecutive elements in period 3 and L forms an anion, J, K and L are Al, Si and P respectively.

A is incorrect. Electrical conductivity of $K > J > L$.
 B is incorrect. Al^{3+} have a larger radius than Si^{4+}
 C is incorrect. Al^{3+} and Si^{4+} are isoelectronic.
 D is correct. PCl_5 would dissolve in water to form an acidic solution which reacts with Mg metal to produce H_2 gas (effervescence).

16	Which solution will form a precipitate when aqueous silver nitrate is added?	
	A	$Na_3[CrCl_6]$
	B	$[Cr(NH_3)_3Cl_3]$
	C	$[Cr(NH_3)_6]Cl_3$
	D	$[Cr(en)(NH_3)_2Cl_2]NO_3$

Answer: C

$[Cr(NH_3)_6]Cl_3$ dissolves in water to form $[Cr(NH_3)_6]^{3+}$ and Cl^- . Cl^- then reacts with Ag^+ to form $AgCl$, a white precipitate.

All other options do not dissolve to give Cl^- .

17	Which statement is not always true for transition metals and its ions?	
	A	The density of transition metals is greater than alkaline metals.
	B	Transition elements or their compounds are widely used as catalyst.
	C	The 3 rd and 4 th ionisation energy of transition metals increases sharply.
	D	The ionic radius of transition metals decreases across the period.

Answer: C

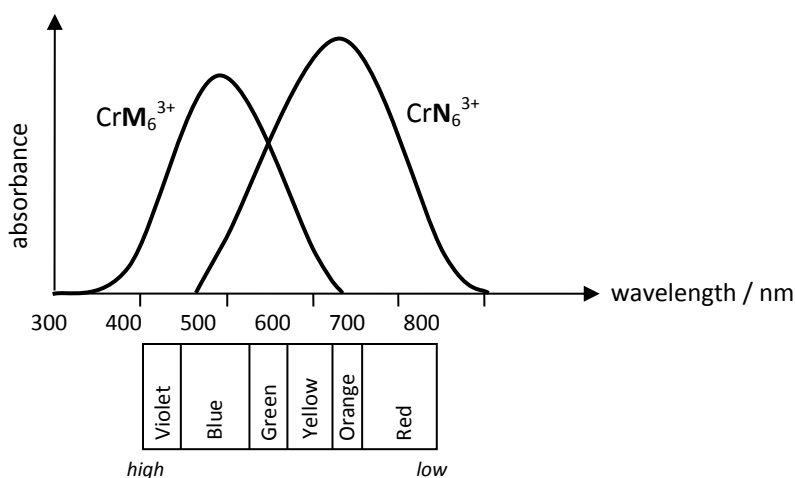
Statement C is not always true. Students can make use of the Data Booklet to see the trend between the 3rd and 4th IE. There is a decrease in the 3rd to 4th IE from Fe to Co.

Statement A is always true as the atomic radius of transition metal are generally smaller than that of the alkaline metal as such there will be more particles per unit volume resulting in greater density for transition metals.

Statement B is always true. Transition elements or their compounds are widely used either as homogenous or heterogeneous catalysts.

Statement D is correct. As we go across the period, the effective nuclear charge increases resulting in stronger forces of attraction between the nucleus and the valence electrons.

- 18** The diagram below shows the visible spectra of two chromium(III) complexes, CrM_6^{3+} and CrN_6^{3+} . The various colours corresponding to the approximate wavelengths in the visible light region are shown below the axis.



What is the colour of each complex and which ligand causes a larger d-orbital splitting?

		colour of CrN_6^{3+}	colour of CrM_6^{3+}	Strength of field ligand
	A	Red	violet	M > N
	B	Red	violet	N > M
	C	Blue	yellow orange	M > N
	D	Blue	yellow orange	N > M

Answer: C

CrM_6^{3+} absorbance is around 450 nm (shorter wavelength, high energy) indicating longer wavelength such as yellow orange/red is transmitted.

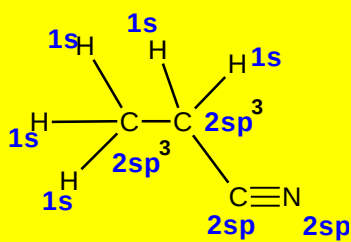
CrN_6^{3+} absorbance is around 750 nm (longer wavelength, low energy) indicating shorter wavelength such as blue/violet is transmitted.

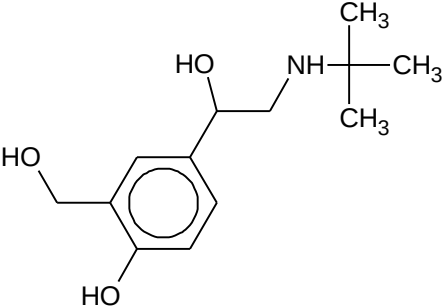
As such strength of ligand $M > N$

19	Covalent bonds are formed by orbital overlap. The shape of unsaturated hydrocarbon molecules can be explained in terms of hybridisation of orbitals. Which bond is not present in $\text{CH}_3\text{CH}_2\text{CN}$?
A	a σ bond formed by $1s - 2sp^3$ overlap
B	a σ bond formed by $2sp - 2sp^2$ overlap
C	a σ bond formed by $2sp^3 - 2sp^3$ overlap
D	a π bond formed by $2p - 2p$ overlap

Answer B

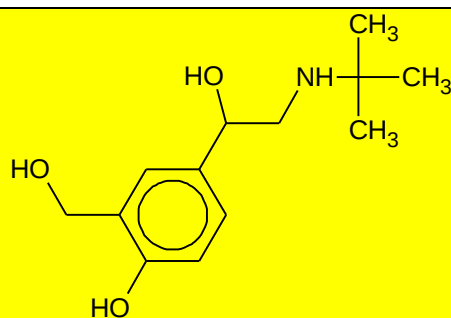
B is incorrect. There is no sp^2 hybridised element in this molecule.



20	Salbutamol is a compound used in inhalers to relieve asthma.  It can be reacted with H_3PO_4 at 250°C . What is the change in the number of stereoisomers after the reaction?		
		Change in number of chiral carbon	Change in number of cis-trans isomers
	A	Decrease of 1	Increase of 2
	B	No change	Increase of 1
	C	Decrease of 1	Increase of 1
	D	No change	Increase of 2

C

When the compound is dehydrated, the following product is formed:



Dehydration

Initial:

Chiral C: 1

Cis – trans: 0

Final:

Chiral C: 0

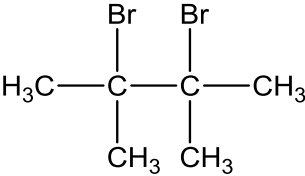
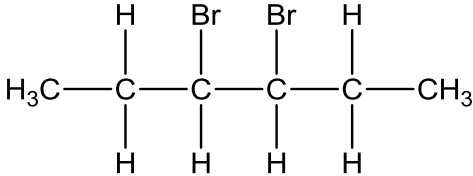
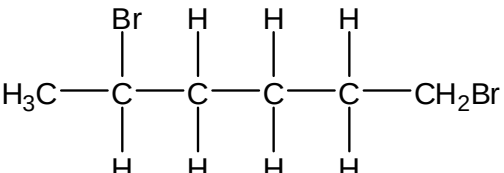
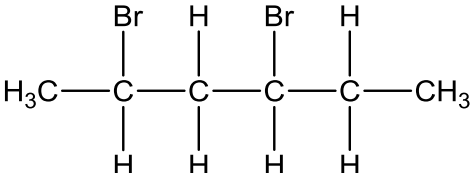
Cis – trans: 1

Change:

Chiral C: decrease of 1

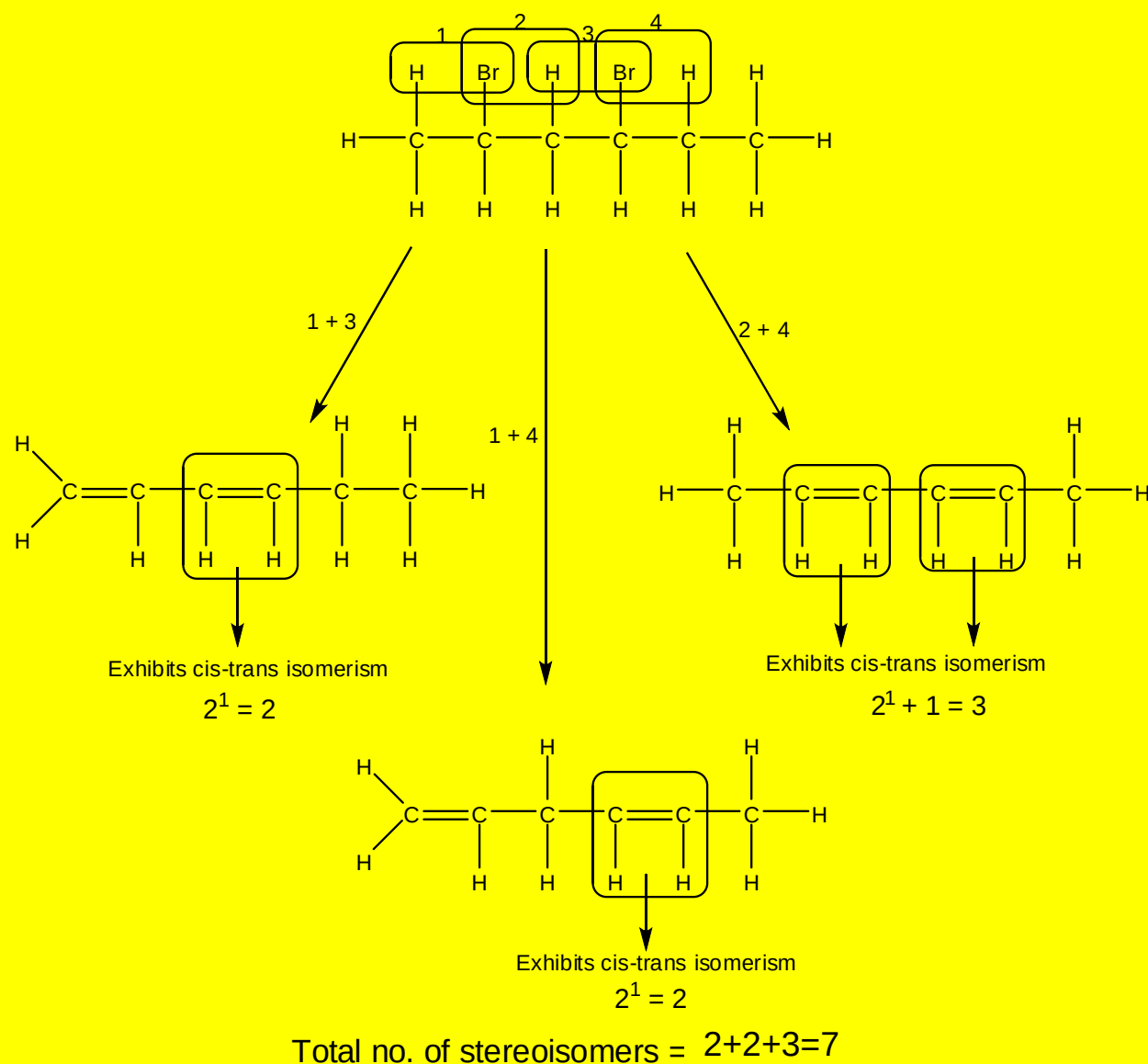
Cis – trans: increase of 1

21	How many structural isomers of carboxylic acids and esters are there in the formula $C_4H_8O_2$?								
	A	3	B	4	C	5	D	6	
Answer: D 6 Structural isomerism includes functional group isomerism, chain isomerism and positional isomerism. $CH_3CH_2CH_2COOH$ $CH_3CH(CH_3)COOH$ $CH_3-C(=O)-O-CH_2CH_3$ $H_3C-O-C(=O)-CH_2CH_3$ $H-C(=O)-O-CH_2CH_2CH_3$ $H-C(=O)-O-CH(CH_3)CH_3$									

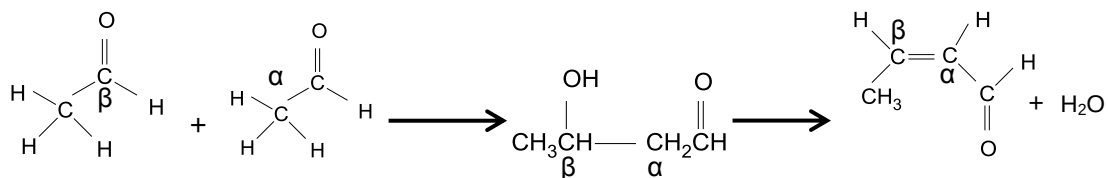
22	Which of the isomers of $C_6H_{12}Br_2$, on reaction with hot ethanolic potassium hydroxide for several hours, gives the greatest number of isomers?				
A			B		
C			D		

Answer : D

The compound undergoes elimination of HBr with hot ethanolic potassium hydroxide.



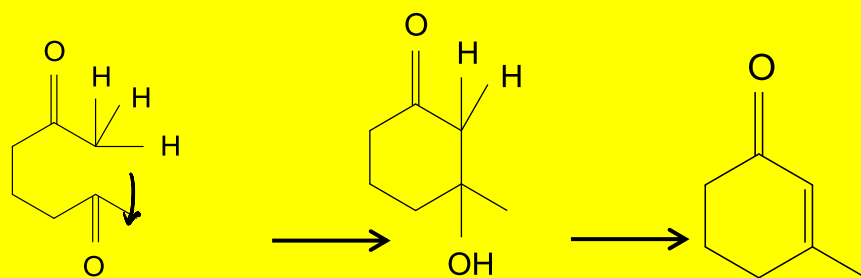
- 23** An aldol condensation is a reaction in which an enol or an enolate ion reacts with a carbonyl compound, followed by dehydration to give a conjugated enone. An example of the reaction can be seen from acetaldehyde below.



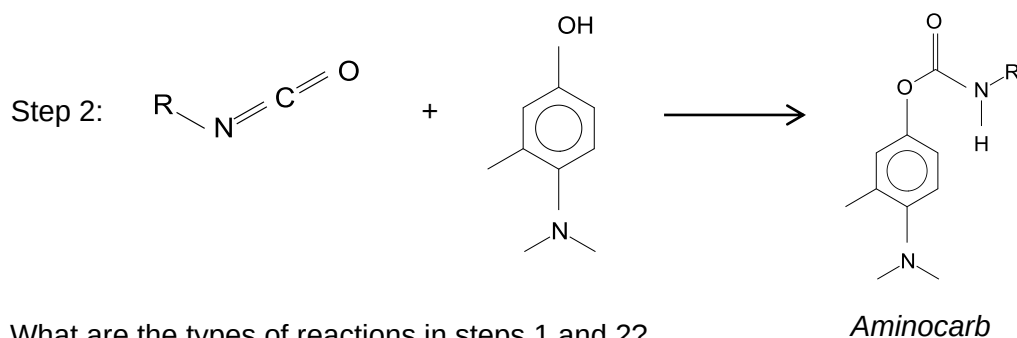
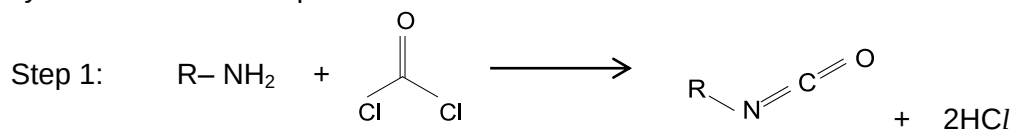
Which one of the following shows the correct product obtained from an aldol condensation?

	Reactants	Product
A		
B		
C		
D		

Answer: B



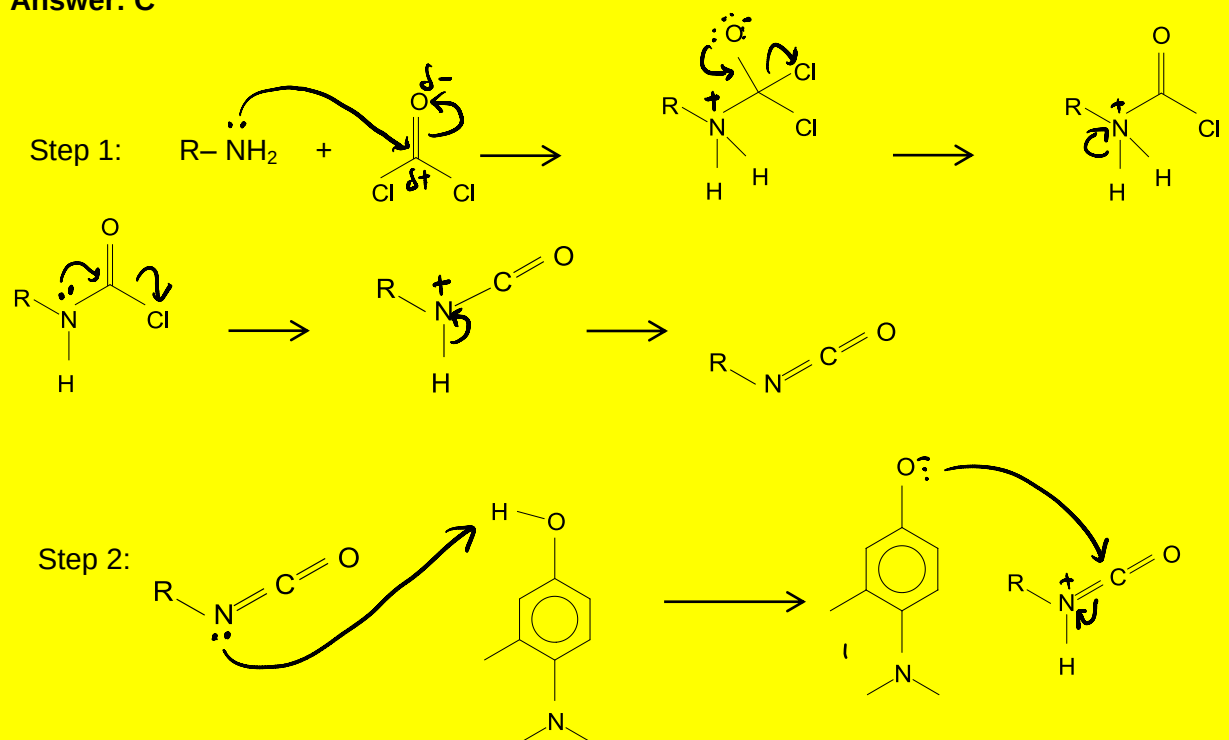
- 24** Aminocarb is an organic compound that is commonly used as a carbamate insecticide to protect cotton fields and forests from insect infestation. Aminocarb can be synthesized in two steps.



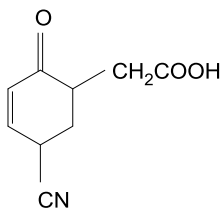
What are the types of reactions in steps 1 and 2?

	Step 1	Step 2
A	condensation	nucleophilic substitution
B	nucleophilic addition	electrophilic addition
C	condensation	nucleophilic addition
D	nucleophilic addition	electrophilic substitution

Answer: C



- 25** Compound **Q** can undergo reduction under suitable conditions. Which of the following shows the correct reducing reagent and the product obtained?



Compound **Q**

	Reducing agent	Product	
A	NaBH_4		
B	LiAlH_4 , dry ether		
C	H_2 , Pt catalyst		
D	H_2 , Ni catalyst		

Answer: A

H_2 , Ni catalyst will cause reduction of ketone, nitrile and alkene.

LiAlH_4 , dry ether will cause reduction of ketone, nitrile and carboxylic acid.

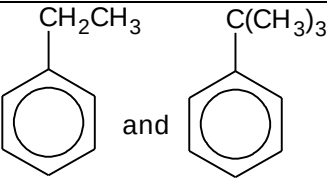
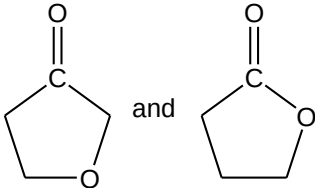
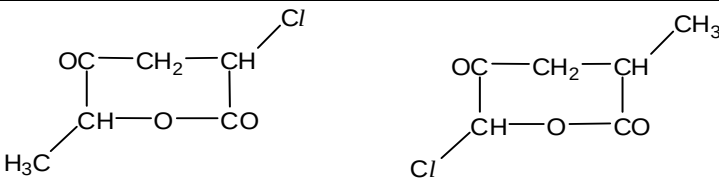
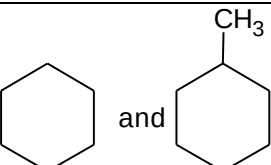
NaBH_4 will cause reduction of ketone only.

26	Which one of the following gives the compounds in order of decreasing acid strength?
A	$\text{CH}_3\text{CH}_2\text{COOH} > \text{CH}_2\text{ClCH}_2\text{COOH} > \text{CH}_3\text{CHClCOOH}$
B	$\text{CH}_2\text{BrCOOH} > \text{CH}_2\text{ClCOOH} > \text{CH}_2\text{FCOOH}$
C	$\text{CF}_3\text{COOH} > \text{CHF}_2\text{COOH} > \text{CH}_2\text{FCOOH}$
D	$\text{CH}_3\text{CH}_2\text{OH} > \text{C}_6\text{H}_5\text{OH} > \text{CH}_3\text{COOH}$

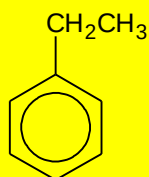
Answer : C

In option C :

F is a very electronegative atom. Since F is electron withdrawing in nature, the more F atoms there are, the more acidic the compound.

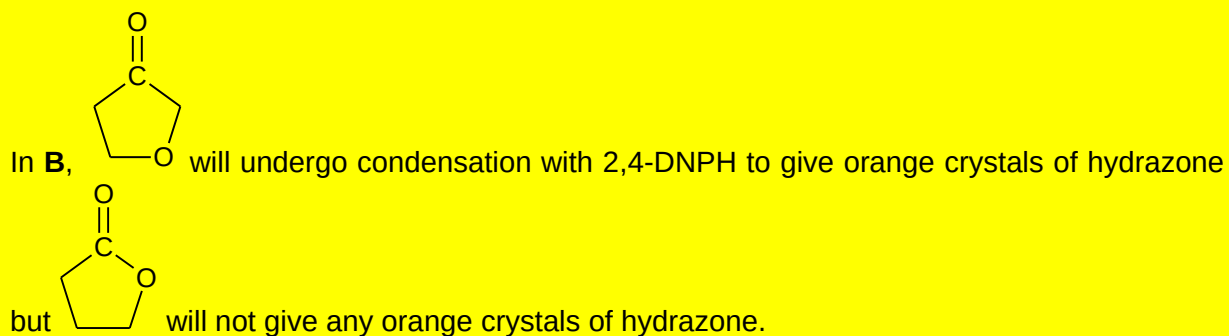
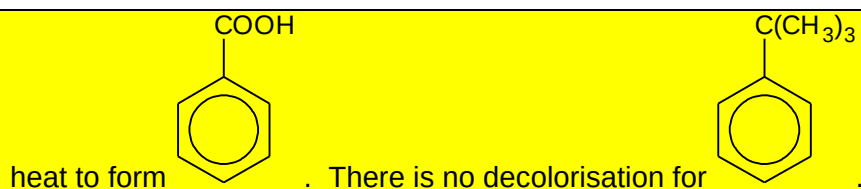
27	Which pair of organic compounds cannot be distinguished by a chemical test?
A	
B	
C	
D	

Answer : D

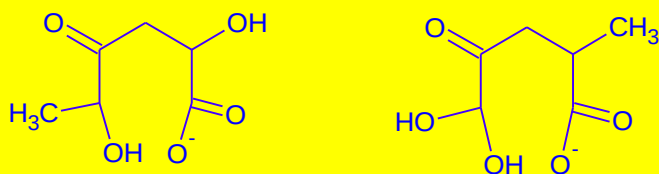


In A, will undergo strong oxidation and decolorise purple KMnO_4 in dilute H_2SO_4 ,

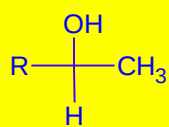
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In **C**, after heating in alkaline condition, hydrolysis of esters and nucleophilic substitution of chloroalkanes take place. The intermediate products are as follow.



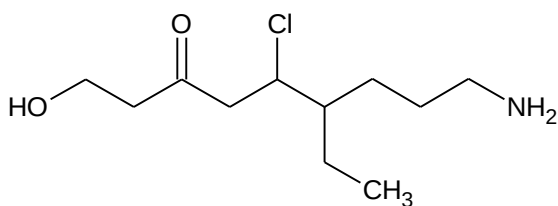
1 will give positive iodoform test since it has the



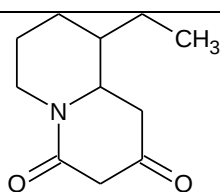
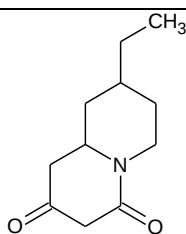
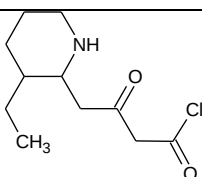
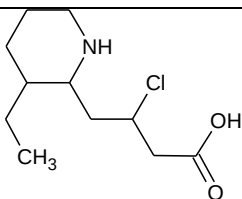
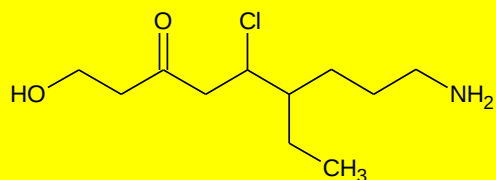
group while **2** will give negative results.

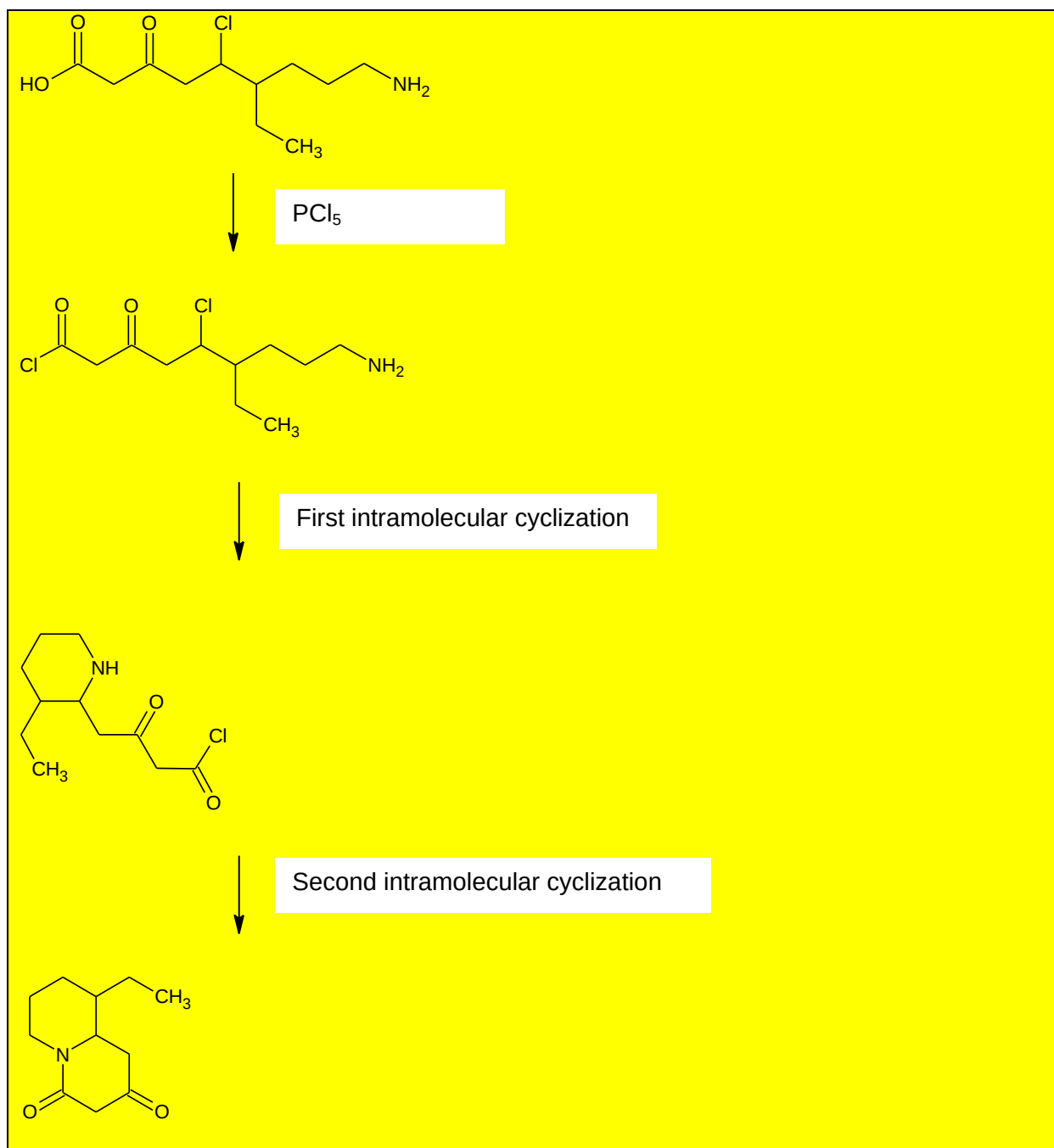
In **D**, both compounds are **cycloalkanes**. There is no chemical test to distinguish them.

28



The compound above is reacted with acidified $K_2Cr_2O_7$, followed by PCl_5 . The resulting compound undergoes intramolecular reactions to form ringed structures under suitable conditions. Which structure is formed?

A**B****C****D****Answer: A**acidified $K_2Cr_2O_7$



30	In the tertiary structure of a water-soluble globular protein extracted from barley grains, it was found that the types of amino acid residues present on the outer surface of the protein were different from those present on the inside. Which of the following amino acids is most likely to be incorporated into the outer surface of such proteins?		
A	$\begin{array}{c} \text{H}_2\text{N}-\text{CH}-\text{CO}_2\text{H} \\ \\ \text{H}_3\text{C}-\text{CH}-\text{OH} \end{array}$	B	$\begin{array}{c} \text{H}_2\text{N}-\text{CH}-\text{CO}_2\text{H} \\ \\ \text{CH}_2-\text{CH}_2-\text{S}-\text{CH}_3 \end{array}$
C	$\begin{array}{c} \text{HN}-\text{CH}-\text{CO}_2\text{H} \\ / \quad \backslash \\ \text{H}_2\text{C} \quad \text{CH}_2 \\ \backslash \quad / \\ \text{CH}_2 \end{array}$	D	$\begin{array}{c} \text{H}_2\text{N}-\text{CH}-\text{CO}_2\text{H} \\ \\ \text{H} \end{array}$

Answer: A

As the protein is "water-soluble", A is the only amino acid with the R-group that may be soluble in water (favourable H-bonds).

For **questions 31 – 40**, one or more of the numbered statements **1** to **3** may be correct. Decide whether each of the statements is or is not correct. The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is to be used as correct response.

31	Which of the following statements about the nitrate(V) ion, NO_3^- , is correct?
1	The hybridisation of the nitrogen atom is sp^2 .
2	NO_3^- has a total of twenty-four valence electrons.
3	All the nitrogen–oxygen bond lengths in NO_3^- are equal in length.
Answer : A (1, 2 and 3 only) Option 1 is correct. The central nitrogen atom is bonded to three oxygen atoms, and has no lone pairs. If you were to draw the Lewis dot structure, one of the nitrogen-oxygen bonds would be a double bond, and the other two would be single bonds, but the three bonds can resonate with each other, so actually it's more like they're all $1\frac{1}{3}$ bonds. Since it's a symmetric atom with three substituents, the geometry around the central atom will be trigonal planar, and the hybridization will be sp^2. Option 2 is correct. Oxygen has 6 valence electrons and there's 3 oxygens. $6 \times 3 = 18$ Nitrogen has 5 valence electrons and there's one nitrogen. $5 \times 1 = 5$ A charge of -1 means you add one electron. +1 $18 + 5 + 1 = 24$ Option 3 is correct. NO_3^- has three resonance structure. One double bond between N and O and another two N - O single bond. Since the double bond can be formed between N and any other O it is said that it has a resonance structure. This means that the true structure of NO_3^- is not any of this structure, but rather a hybrid of all three. Hence all 3 bond lengths would be identical and shorter than single bond as it is basically an average between single and double bond.	

32	When an equilibrium is established in a reversible reaction, the standard Gibbs free energy, $\Delta G^{\ominus}_{\text{eq}}$, is related to the equilibrium constant, K_{eq}, by the following equation. $\Delta G^{\ominus}_{\text{eq}} = -2.303 RT \log K_{\text{eq}}$ where R is the gas constant and T is the temperature in Kelvin. Which of the following statements are correct?	
	1	At constant temperature, a shift in position of equilibrium to the right results in the same value of $\Delta G^{\ominus}_{\text{eq}}$.
	2	The reaction is spontaneous for all values of K_{eq} .
	3	Adding a catalyst makes $\Delta G^{\ominus}_{\text{eq}}$ more negative.
Answer : D (1 only) Option 1 is correct. If temperature is constant, there is no impact on value of K_{eqm}. K_{eqm} is temperature dependent. Hence, no impact on $\Delta G^{\ominus}_{\text{eq}}$. Option 2 is wrong. The spontaneity of the reaction depends on the K_{eqm} of the reaction. So it may not be spontaneous at all temperature. Option 3 is wrong. Catalyst only increases the rate of forward reaction and has no impact on $\Delta G^{\ominus}_{\text{eq}}$.		

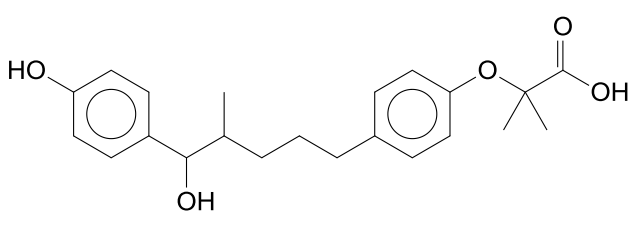
33	The mechanism for a given reaction is shown below. $ \begin{array}{ll} 2\text{S} \rightleftharpoons \text{U} & \text{fast} \\ \text{U} + \text{T} \rightarrow \text{V} & \text{slow} \\ \text{V} + \text{T} \rightarrow \text{S}_2\text{T}_2 & \text{fast} \end{array} $ What conclusions can be drawn from the information given above?	
	1	Rate = $k[\text{U}][\text{T}]$
	2	The overall equation is $2\text{S} + 2\text{T} \rightarrow \text{S}_2\text{T}_2$.
	3	The rate of formation of S_2T_2 is proportional to the initial concentration of T .
Answer: C (2 and 3 only) Option 1 is incorrect. Rate = $k'[\text{S}]^2[\text{T}]$ where $k' = kK_c[\text{U}]$ Option 2 is correct. Option 3 is correct. The rate of formation of S_2T_2 is proportional to [T]		

34	The numerical values of the solubility product of calcium carbonate and calcium fluoride are 8.0×10^{-7} and 4.5×10^{-11} respectively at 30°C . Which of the following statements are correct?	
	1	Calcium carbonate is more soluble than calcium fluoride.
	2	Addition of calcium nitrate solution to a solution containing equal concentration of carbonate and fluoride ions causes calcium carbonate to precipitate out first.
	3	The solubility product of calcium carbonate decreases in a solution containing calcium nitrate.
	Answer: D (1 only) $\underset{\text{s}}{\text{CaCO}_3(\text{s})} \rightleftharpoons \underset{\text{s}}{\text{Ca}^{2+}(\text{aq})} + \underset{\text{s}}{\text{CO}_3^{2-}(\text{aq})}$ $K_{\text{sp}} = [\text{Ca}^{2+}][\text{CO}_3^{2-}]$$8.0 \times 10^{-7} = (\text{s})(\text{s})$$\text{s} = 8.94 \times 10^{-4} \text{ mol dm}^{-3}$ $\underset{\text{s}}{\text{CaF}_2(\text{s})} \rightleftharpoons \underset{\text{s}}{\text{Ca}^{2+}(\text{aq})} + \underset{2\text{s}}{2\text{F}^{-}(\text{aq})}$ $K_{\text{sp}} = [\text{Ca}^{2+}][\text{F}^{-}]^2$$4.5 \times 10^{-11} = (\text{s})(2\text{s})^2$$\text{s} = 2.24 \times 10^{-4} \text{ mol dm}^{-3}$	
	From the above calculation, calcium carbonate is more soluble than calcium fluoride. Thus Option 1 is correct. The calculation above is necessary as the solubility product of the two salt cannot be compared directly as their K_{sp} expression is different. On addition of $\text{Ca}(\text{NO}_3)_2$ to equal concentration of CO_3^{2-} and F^{-}, calcium fluoride will precipitate out first. Option 3 is wrong as the solubility product is dependent only on temperature change.	

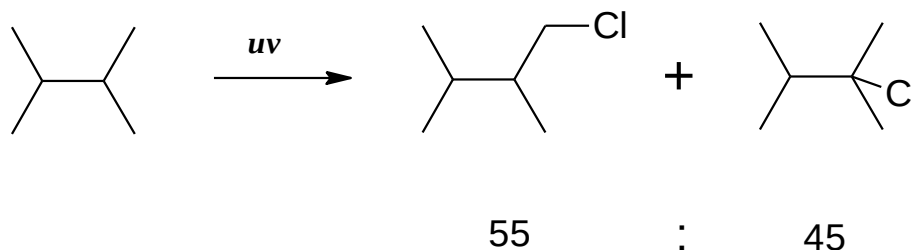
35	When drops of sodium hydroxide were added to a green solution of chromium(III) nitrate, a grey-green precipitate was formed. The precipitate dissolved when excess sodium hydroxide was added, forming a dark green solution. Subsequent additions of liquid ammonia caused the solution to turn violet. According to the information given above, which of the following statements is correct?	
1	Hydroxide ions act as ligands in the first addition of sodium hydroxide.	
2	When liquid ammonia is added, ligand exchange occurs.	
3	The equilibrium constant, K_{stab} for the formation of the ammonia-containing complex is larger than that of the hydroxide-containing complex.	
Answer: C (2 & 3) Option 1 is wrong. When the first few drops of NaOH were added, the hydroxide is not acting as the ligand in this precipitation reaction. Option 2 is correct. When liquid ammonia is added, colour change is detected indicating that ligand exchange reaction has occurred.		

	Option 3 is correct. Since the ammonia-containing complex is favourable as compared to the hydroxide-containing complex, the equilibria constant will be larger. A more stable complex has a higher K_{stab} value.
--	---

36	A catalytic converter is part of the exhaust system of modern cars. Which reactions occur in a catalytic converter?
1	$C_xH_y + (x + y/4)O_2 \rightarrow xCO_2 + (y/2) H_2O$
2	$2CO + 2NO \rightarrow 2CO_2 + N_2$
3	$2SO_2 + 2NO \rightarrow 2SO_3 + N_2$
	Answer: B (1 and 2 only) The objective of catalytic convertor is to remove the harmful component in the exhaust. Statement 1 removes the unburnt hydrocarbons Statement 2 removes the carbon monoxide and oxides of nitrogen Statement 3 No sulfur-containing product in the exhaust gas

37	 Z Which of the following statements about the reactions of Z is incorrect?
1	1 mol of Z reacts with $SOCl_2$ to release 3 mol of HCl gas.
2	1 mol of Z reacts with Na metal to release 3 mol of H_2 gas.
3	1 mol of Z reacts with Na_2CO_3 to release 2 mol of CO_2 gas.
	Answer: A (1, 2 and 3 only) 1 $SOCl_2 \equiv 1 -OH$ group $\equiv 1 HCl$ Only ROH and $-COOH$ group reacts with 2 mol $SOCl_2$, producing 2 mol HCl gas. (*phenol does not react with $SOCl_2$) 1 Na $\equiv 1 -OH$ group $\equiv \frac{1}{2} H_2$ Thus, 1 mol of Z reactions with 3 mol Na, producing $\frac{3}{2}$ mol H_2 gas. $2RCOOH + Na_2CO_3 \rightarrow 2RCOO^-Na^+ + CO_2 + H_2O$ Only 1 $-COOH$ group present in Z, producing $\frac{1}{2}$ mol of CO_2 gas.

- 38** 2,3-dimethylbutane reacts with chlorine under uv to yield two monochlorinated products as shown in the reaction below.



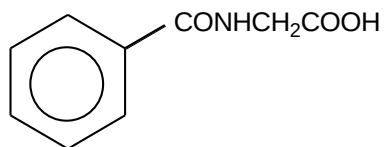
The product ratio was given to be 55 : 45 as illustrated above.

Which of the following statements explain the resultant product ratio?

- | | |
|----------|---|
| 1 | There are more primary hydrogen atoms than tertiary hydrogen atoms. |
| 2 | The intermediate tertiary radical is more stable. |
| 3 | $H\bullet$ is not formed in the reaction mechanism. |

Answer : B (1 and 2 only)

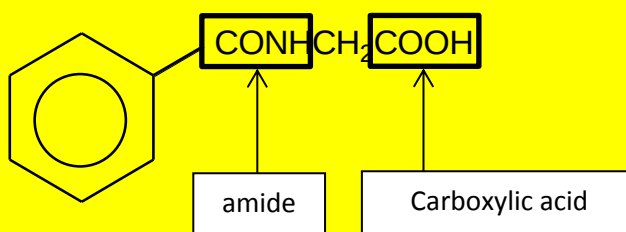
39 Benzoylglycine (hippuric acid) was first isolated from a urine sample.



Which properties does this compound possess?

- | | |
|----------|---|
| 1 | It can be hydrolysed to produce an amino acid. |
| 2 | It can be made by reacting benzoyl chloride with aminoethanoic acid. |
| 3 | It can be neutralised by reaction with cold aqueous sodium hydroxide. |

Answer : A (1, 2 and 3 only)

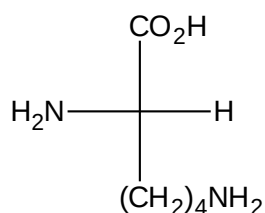


1 is correct as amides can be hydrolysed at the -CONH- bond to produce glycine, an amino acid.

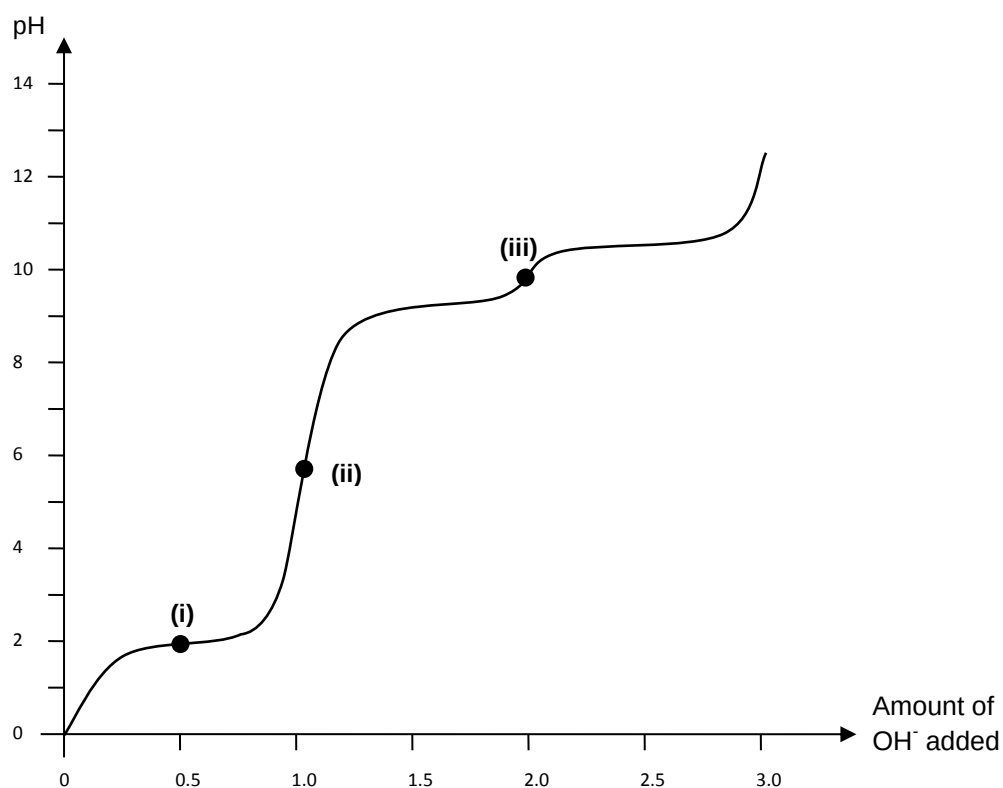
2 is correct. To make a secondary amide, the starting materials must be an acyl chloride and the primary amine.

3 is correct. The carboxylic acid can be neutralised easily to form a carboxylate salt.

- 40** Lysine is an essential amino acid found in the body. It has three pK_a values associated with it: 2.2, 9.0 and 10.5.



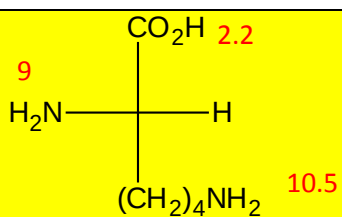
When one mole of protonated lysine was titrated against hydroxide ions, the following pH curve is obtained:



Which of the following statements are true with respect to the curve above?

- | | |
|----------|--|
| 1 | The major species present at point (iii) has no net charge. |
| 2 | The major species present at point (ii) will migrate towards the cathode of an electrolytic cell. |
| 3 | Equal amounts of $\text{H}_3\text{N}^+\text{CH}(\text{CO}_2\text{H})(\text{CH}_2)_4\text{NH}_3^+$ and $\text{H}_3\text{N}^+\text{CH}(\text{CO}_2\text{H})(\text{CH}_2)_4\text{NH}_2$ are present at point (i) . |

Answer: B (1 and 2 only)



Point A: max buffering capacity

Species present: $\text{H}_3\text{N}^+\text{CH}(\text{CO}_2\text{H})(\text{CH}_2)_4\text{NH}_3^+$ and $\text{H}_3\text{N}^+\text{CH}(\text{CO}_2^-)(\text{CH}_2)_4\text{NH}_3^+$

Point C: 2nd equivalence point

Species: $\text{H}_2\text{NCH}(\text{CO}_2^-)(\text{CH}_2)_4\text{NH}_3^+$

Net charge: 0

Point B: first equivalence point

Species: $\text{H}_3\text{N}^+\text{CH}(\text{CO}_2^-)(\text{CH}_2)_4\text{NH}_3^+$

Net charge: +1 (migrate towards cathode)

END OF PAPER 1

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